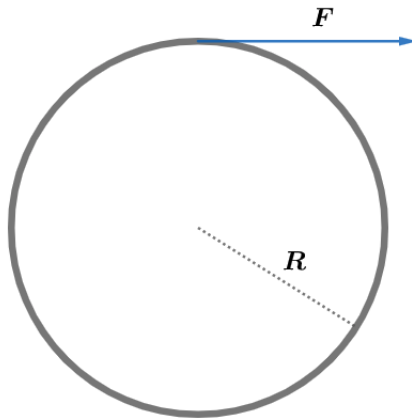


2007 F=ma Exam: Problem 10

Kevin S. Huang



Recall for rotational motion,

$$\tau = I\alpha$$

We are given the moment of inertia of the wheel:

$$I = mR^2$$

The force is applied perpendicular to the moment arm:

$$\tau = FR$$

Thus,

$$FR = mR^2\alpha$$

$$F = mR\alpha$$

Both wheels have the same mass, so for both to have the same angular acceleration,

$$F \propto R$$

Hence,

$$R_2 = 2R_1$$

$$F_2 = 2F_1 = 2\text{ N}$$

so the answer is D.